School of Engineering and Applied Science

Lecture Scribe: Lecture 22 (CSE 400)

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1) List of Topics Covered

1. **Chebyshev's Inequality**
   - Intuition and Key Idea
   - Proof using Markov's Inequality
2. **Running Example: Coin Flipping**
   - Setup and Binomial Distribution application
   - Comparison with Markov's Bound
3. **Real-Life Applications**
   - Exam Scores Scenario
   - Network Latency SLA

2) Explanation of Topics Covered

1. **Chebyshev's Inequality**
   Chebyshev's Inequality is a powerful tool in probability that uses both the mean (μ) and variance ($Var(X)$) to control the deviation of a random variable from its mean. It helps bound the probability of "tail regions"—values that are far from the average. Unlike Markov's inequality, which only requires the mean and applies only to non-negative variables, Chebyshev's bound often provides a tighter estimate because it incorporates the spread (variance) of the data.

2. **Intuition**
   Geometrically, it limits how much area can exist in the tails of a distribution at a distance 'a' from the mean. As the sample size or certain parameters increase, these bounds typically decrease, showing that values are more likely to stay near the mean.

3) List of Definitions and Theorems

1. **Definition: Markov's Inequality**
   For a non-negative random variable X and any $a > 0$:
$$P(X \geq a) \leq \frac{E[X]}{a}$$

2. **Theorem Statement: Chebyshev's Inequality**
   For a random variable X with mean μ and variance σ^2, for any $a > 0$:
$$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

3. **Detailed Step-by-Step Proof**
   1. Consider the event $|X - \mu| \geq a$.
   2. Square both sides to relate it to variance: $(X - \mu)^2 \geq a^2$.
   3. Apply **Markov's Inequality** to the non-negative random variable $(X - \mu)^2$:
$$P((X - \mu)^2 \geq a^2) \leq \frac{E[(X - \mu)^2]}{a^2}$$
   4. Substitute the definition of variance $Var(X) = E[(X - \mu)^2]$:
$$P(|X - \mu| \geq a) \leq \frac{Var(X)}{a^2}$$

4) Important Examples

Example 1: Exam Grades with Variance

Problem : The average grade is 70% and the standard deviation is 5%. The "A" cutoff is 90%. Apply Chebyshev's inequality to obtain a tighter upper bound on the fraction of "A" students.

Solution:

- **Step 1:** Identify mean and variance.
  $\mu = 70$, $\sigma = 5 \Rightarrow \sigma^2 = 25$.

- **Step 2:** Define the event.
  An "A" student has $X \geq 90$. Note that $X \geq 90 \Rightarrow |X - 70| \geq 20$. We use the two-sided Chebyshev bound to cover this one-sided event.

- **Step 3:** State Chebyshev with $a = 20$.
  $P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$.

- **Step 4:** Substitute values.
  $P(X \geq 90) \leq P(|X - 70| \geq 20) \leq \frac{25}{20^2} = \frac{25}{400}$.

- **Result:** $P(X \geq 90) \leq 0.0625$ (or $\frac{1}{16}$). This is much tighter than the bound provided by Markov's inequality (0.778).

Example 2: Network Latency SLA

Problem : A network route has average latency $E[X] = 50$ ms and variance $Var(X) = 16\ \text{ms}^2$. Bound the probability that a packet's latency falls **outside** the range $[30\ \text{ms}, 70\ \text{ms}]$.

Solution:

- **Step 1:** Identify given parameters.
  $\mu = 50$, $\sigma^2 = 16$.
- **Step 2:** Rewrite failure condition as absolute deviation.
  $X \notin [30, 70] \iff |X - 50| \geq 20$.
- **Step 3:** Apply Chebyshev with $a = 20$.
  $P(|X - 50| \geq 20) \leq \frac{16}{20^2} = \frac{16}{400}$.
- **Result:** $P(X \text{ is outside range}) \leq 0.04$.

5) List of Important Formulas

Formula / Concept	Result
Markov's Inequality	$P(X \geq a) \leq \frac{E[X]}{a}$
Chebyshev's Inequality	$P(\lvert X - \mu \rvert \geq a) \leq \frac{Var(X)}{a^2}$
Binomial Mean (Fair Coin)	$\mu = \frac{n}{2}$
Binomial Variance (Fair Coin)	$Var(X) = \frac{n}{4}$